Bahria University Lahore Campus

Department of Computer Sciences

Course Code/number	**CSC-114**
Course Title/Name	**Introduction To Information & Communication Technology**
Credit Hours/Contact hours	2+1
Degree Program	Bachelor of Science in Computer Science (BSCS)
Prerequisites or Co-requisites	None
Assessment Methods and Weightage	Quizzes 10 Assignments/Projects 20 Mid-Term Examination 20 Final Examination 50 **Total 100**
Textbook (or **Laboratory Manual** for Laboratory Courses)	▪ Introduction to Computers by peter Norton 7th Edition
Reference Material (With Edition, ISBN#)	▪ Discovering Computers by Misty E. Vermaat, Susan I, Sebok, Sleven M Freund ▪ Fundamentals of Information Technology by Alexis Leon & Mathews Leon, Leon Vikas ▪ Using Information Technology: A Practical Introduction to Computer & Communications, by Williams, 6th Edition ▪ Understanding Computers: Today and Tomorrow, Comprehensive, 15 th Edition, Cengage Learning, Nell Dale, John Lewis ▪ Computer Science Illuminated, 5th Edition, Jones & Bartlett Learning.
Web Resources/ URL (if any)	

Instructor Name/Cluster Head/Subject Expert	Instructor Name: Mr. Taimoor Aamer Chughtai Designation: Assistant Professor Cluster Head Name: Dr. Irum Noreen
Course Aims:	The objectives of this course include making the students learnthe fundamental concepts of computing technologies and enable them to apply the concepts of number systems, programming, and databases. Furthermore, understanding networking and internet technologies also makes part of this course

Course Learning Outcomes

CLO	Goal	PLO	BT
1	Describe the components of a computer system	1	C1
2	Understand the fundamentals of operating systems, databases, and number system	2	C2
3	Explain Networking and Internet Technologies	2	C2

Course Description/Catalogue	Computer systems, binary number system, data representation, computer language, data processing, data communication, networking.

Lecture Plan (16 Weeks)

Week # and Date	Lecture/ Hour #	Topic to be covered	Learning outcomes	Reference Text
1	2	Introduction, course objectives, broad course outline, topics covered in the semester, schedule of assignments, quizzes, and project etc. Marks and grade policy. Plagiarism restrictions. Attendance.	Introduction to the course	Handouts
2	2	Define the term, computer. Discuss the advantages and limitations of computer. Brief history of computers. Categories of computers. Examples of computer usage.	Learn about definition of computer, its advantages and limitations, usage of computer in various fields of life, brief history of computer, different Categories of computers, different part of	Lecture Handouts, Chapter 1A, B of text book

			Discuss different parts of System Unit. Types and function of processors, Types of memory, input devices and output devices	system Unit, types and function of processors, types of memory, input devices and output devices.	
	3	2	Interacting with computers: Keyboard Functions and types, working of keyboard, Types of Mouse, and working of mouse, Repetitive stress injuries. Inputting data in other ways. Devices for hand, Touch screens, Optical input devices, Optical character recognition, Audio visual input devices, microphone, digital camera	Learn how the computers and human interacts, types of keyboard, functions and working of keyboard, types of mouse, and working of mouse, entering data to computers in other ways like touch screen, optical input devices, how instruction can be given through microphone & digital camera.	Lecture Handouts, Chapter 2A, 2B of text book
	4	2	Video and sound, Monitors, video cards, sound system, Printing, types of printers,	Understand the types of audio and video devices. Learn about printers their types and how to compare printer on the basis of different parameters	Lecture Handouts, Chapter 3A, 3B of text book
	5	2	Introduction to Number Systems. Conversion of decimal numbers to binary, octal and hexadecimal numbers. Conversion of binary to decimal, octal and hexadecimal numbers. Conversion of octal to hexadecimal numbers. Conversion of hexadecimal numbers to octal numbers.	To know about different number systems. Learn conversion of decimal numbers to binary, decimal to octal and decimal to hexadecimal numbers, conversion of binary to decimal, octal and hexadecimal numbers, conversion of octal to hexadecimal numbers, conversion of hexadecimal numbers to octal numbers.	Handouts. Chapter 4A of text book.

	6	2	Binary Arithmetic. Binary Addition rules. Examples of addition of binary numbers Examples of multiplication of binary numbers.	Learn how to add binary numbers. Learn how to multiply binary numbers. Learn how to divide binary numbers.	Handouts. Chapter 4A of text book.
	7	2	Binary Subtraction rules. Subtraction with 1's complement and 2's complement method. Examples of subtraction of binary numbers. Examples of division of binary numbers.	Learn how to subtract binary numbers with 1's complement and 2's complement methods.	Handouts. Chapter 4A of text book
	8	2	REVISION	REVISION	
	Mid Term Exam				
	9	2	Introduction to file management system. Definition of database and database management system. File management system vs database management system.	Learn about file management system. Learn about database and database management system. Differentiate between File management system and database management system.	Handouts and chapter 11A,B of text book.
	10	2	Web Technologies (HTML)	Learn about different web technologies	Handouts and Chapter 8A of text Book.
	11	2	Web Technologies(CSS)	Learn about different web technologies	Handouts and Chapter 8A of text Book.
	12	2	Types of application software. Steps to develop custom software. Types of acquiring software. Software for small business	Learn about different type of application software and steps in developing custom software, different types of acquiring software, and	Handouts

				different software for small business.	
	13	2	Operating System	Overview of operating system, Basics and its purpose, Types of operating system, Survey of PC and Network Operating system	Handouts and Chapter 6A and 6B of text Book.
	14	2	Security and privacy Cryptography	Learn different encryption algorithms and security goals	Handouts
	15	2	Introduction to computer networks and security. Types of networks. Network Topologies: Bus, Star, Ring, Mesh. Pros and Cons of each topology.	Learn about basic networking concepts and security. Learn about different network topologies.	Handouts, Chapter 7A of text book
	16	2	Software programming and development	Understand computer program and hardware, software interaction. Learn how program solve problems	Handouts
Final Term Exam					

Semester Calendar for Assignments/Quizzes/Project

Assignments/ Projects and quizzes Plan	Week #	Assignment No.	Quiz No.	Project Title	Assignment/Project Quiz Date	Result Date of Assignment/Project/ Quiz
	1					
	2					
	3	Assignment-1				
	4		Quiz-1			
	5				Assignment-1	Quiz-1
	6					Assignment-1
	7	Assignment-2	Quiz-2			

	8					Quiz-2
	9				Assignment-2	
	10					Assignment-2
	11					
	12					
	13					
	14	Assignment-3				
	15					Assignment-3
	16					

Instructor Name: **Taimoor Aamer Chughtai**

Instructor Signature _____

Date _____